Fundamental Theorem of Linear Programming

Math Camp 2026 — Teaching Notes

On a nonempty polyhedron, a bounded linear objective attains its optimum.
The geometry: a sliding hyperplane must touch a flat face.

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1 Statement

Linear programs ask for the best value of a linear function over a polyhedron. The fundamental theorem says that if the value is finite, an optimizer actually exists.

Theorem 1 (Fundamental theorem of linear programming). *Let $P = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} \le \mathbf{b}\}$ be a nonempty polyhedron and let $f(\mathbf{x}) = \mathbf{c}^\mathsf{T}\mathbf{x}$. If f is bounded above on P, i.e. $\sup_{\mathbf{x}\in P} \mathbf{c}^\mathsf{T}\mathbf{x} < \infty$, then f attains its maximum on P: there exists $\mathbf{x}^* \in P$ with $\mathbf{c}^\mathsf{T}\mathbf{x}^* = \sup_{\mathbf{x}\in P} \mathbf{c}^\mathsf{T}\mathbf{x}$.*

The same statement holds for a minimum when f is bounded below (apply the theorem to $-f$). We focus on maximisation below.

Remark 2 (This fails for general closed sets). *Closedness and a finite supremum are not enough. Take $P = \{(x, y) \in \mathbb{R}^2 : y \ge e^x\}$ and $f(x, y) = -y$. Then $f \le 0$ on P, so $\sup f = 0$, but no point in P has $f = 0$. The set approaches the horizontal line $y = 0$ asymptotically; its boundary is curved. Polyhedra are different: finitely many flat faces, no such asymptotic gaps.*

Economics. In LP applications (profit maximisation, cost minimisation), "the optimum value is M" is only useful if some feasible plan actually achieves M. This theorem guarantees that when the problem is well posed (nonempty feasible set, finite optimum), an optimal solution exists.

The proof is geometric: slide a family of parallel hyperplanes until the last one still touches P.

2 Geometric proof: the sliding hyperplane

Fix $\mathbf{c} \ne \mathbf{0}$ and write $M = \sup_{\mathbf{x}\in P} \mathbf{c}^\mathsf{T}\mathbf{x}$. Consider the parallel hyperplanes

$$H_t = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{c}^\mathsf{T}\mathbf{x} = t\}, \qquad t \in \mathbb{R}.$$

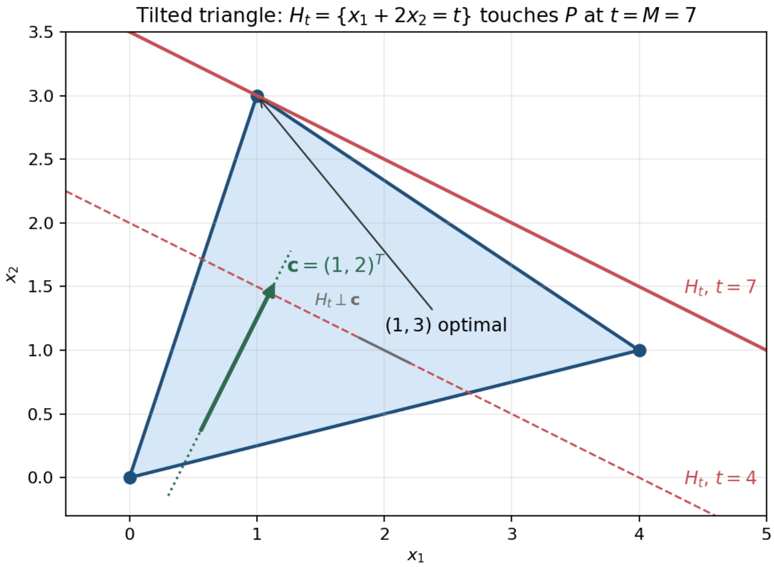
As t increases, H_t moves in the direction of $\mathbf{c}$. Since M is finite, H_t misses P for all $t > M$, and H_t meets P for some $t \le M$. The claim is that the highest touching hyperplane is H_M itself.

Key idea. Project P onto the line spanned by $\mathbf{c}$. For a polyhedron, this projection is a **closed** interval with right endpoint M; hence M is attained.

Claim 1 (Closed projection). *Let $S = \{\mathbf{c}^\mathsf{T}\mathbf{x} : \mathbf{x} \in P\} \subset \mathbb{R}$. Then S is a closed interval of the form $(-\infty, M]$ or $[a, M]$ for some $a \le M$. In particular, $M \in S$.*

Proof sketch. A polyhedron is an intersection of finitely many halfspaces, so P is closed and convex. The linear map $\mathbf{x} \mapsto \mathbf{c}^\mathsf{T}\mathbf{x}$ sends P to a closed convex subset of $\mathbb{R}$, hence an interval. Since M is the supremum of S, the interval is bounded above with right endpoint M. Closedness of the image forces $M \in S$: otherwise $S \subset (-\infty, M)$, contradicting that M is the *least* upper bound. Any $\mathbf{x}^* \in P$ with $\mathbf{c}^\mathsf{T}\mathbf{x}^* = M$ is optimal. □

Why polyhedra are special (2D picture). Rotate coordinates so $\mathbf{c} = (1, 0)^\mathsf{T}$; then $f(\mathbf{x}) = x_1$ and H_t is the vertical line $x_1 = t$. The projection S is the set of first coordinates of points in P. A polygon cannot approach $x_1 = M$ asymptotically: boundary edges are straight segments. A line either crosses $x_1 = M$ (impossible if M is the supremum), lies on $x_1 = M$ (a vertical face, so M is attained), or turns away earlier (the rightmost point is then a vertex). So the right boundary is built from finitely many closed edges and vertices, and $P \cap H_M \ne \emptyset$.



Example (tilted triangle, tilted objective). Let P be the triangle with vertices $(0, 0)$, $(4, 1)$, and $(1, 3)$, and take $\mathbf{c} = (1, 2)^\mathsf{T}$, so $f(\mathbf{x}) = x_1 + 2x_2$. The hyperplanes $H_t = \{\mathbf{x} : x_1 + 2x_2 = t\}$ are *tilted* lines (not vertical). Evaluating at the vertices gives $f = 0$, 6, and 7, so $M = 7$ and the maximum is attained at $(1, 3)$. The highest hyperplane H_7 touches P at that vertex; for $t > 7$ the line misses the triangle entirely.

Remark 3 (Unbounded polyhedra)**.** *P may be unbounded (e.g. a wedge opening to the left). If f is still bounded above, no recession direction of P can increase $\mathbf{c}^\mathsf{T}\mathbf{x}$. The right "cap" is still cut out by finitely many bounding hyperplanes, so the same closed-projection argument applies.*

Key idea. Suprema are attained on polyhedra because boundaries are **flat**, not curved. The sliding hyperplane stops on a face; that face contains an optimizer.